

Conv-ChArT — loss formulation

Penalty-reduced focal on a Gaussian target, shared across both heads

The target

Per-corner Gaussians on the full-resolution grid, **combined by max** (never sum, per CornerNet), with the peak pixel forced to exactly 1.0:

$$Y(u) = \max_{i \in \text{visible}} \exp\left(-\frac{\|u - p_i\|^2}{2\sigma^2}\right), \quad Y(\text{rint}(p_i)) \equiv 1$$

- p_i — the i -th ground-truth corner, at **sub-pixel** float precision.
- $\sigma_{\text{hm}} = 2.0$ px at full resolution; $\sigma_{\text{cls}} = 1.0$ at H/4, where the cell coordinate is $x_c = (x + 0.5)/4 - 0.5$.
- **Max, not sum**: two nearby corners must remain two peaks. Summing would merge them into one taller blob and destroy the very separation the decoder is asked to find.
- **Forcing the peak to 1.0** matters: without it the $Y = 1$ branch of the loss below is never reachable, because a Gaussian sampled at a non-integer centre never attains 1.
- The target is **not one-hot**. A near-miss is scored by how near it is.

The loss

Penalty-reduced focal (CornerNet), evaluated in **logit space** — the naive $\log(1 - \hat{p})$ form NaNs under bf16 near the forced peaks. With $\hat{p}_u = \sigma(z_u)$:

$$\mathcal{L} = - \sum_u \begin{cases} (1 - \hat{p}_u)^\alpha \log \hat{p}_u & Y_u = 1 \\ (1 - Y_u)^\beta \hat{p}_u^\alpha \log(1 - \hat{p}_u) & \text{otherwise} \end{cases}$$

$$\mathcal{L}_{\text{total}} = \frac{\mathcal{L}_{\text{hm}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}}{\max(\sum_{\text{batch}} N_{\text{vis}}, 1)}$$

Parameters

- $\alpha = 2$ — the **focal** term. On a 640×480 map with 16 positives, 0.005% of pixels are positive; without it the loss is dominated by already-correct background and collapses to predicting “no corner” everywhere.
- $\beta = 4$ — the **penalty reduction**. A negative adjacent to a corner has $Y \approx 0.88$, so $(1 - Y)^4 \approx 2 \times 10^{-4}$: it is barely punished. **This is the only place the Gaussian’s graded values enter the loss** — the positive set is $Y = 1$ exactly, identical to a one-hot target’s.
- $\lambda_{\text{cls}} = 1.0$ — both heads carry exactly 307,200 elements at the trained variant, so equal weighting is derived, not tuned.
- N_{vis} — **one shared normaliser** across both heads: the batch total of visible corners. A per-channel $N \in \{0, 1\}$ would inflate $\mathcal{L}_{\text{cls}}$ by roughly 16 $\times$. Clamped at 1 so an all-negative batch ($N = 0$) needs no special case — CornerNet’s own convention.
- Final logit-conv biases initialise to -2.19 ($\pi = 0.1$), so training starts from “probably not a corner” rather than from an uninformative 0.5.

The refiner’s loss

Same functional form on a 64×64 target at $8\times$ over the crop’s central 8×8 px, $\sigma = 1.5$, peak forced at $\text{rint}(u^*)$:

$$u^* = 31.5 + 8d, \quad d \in [-3.9375, 3.9375] \text{ px}$$

Decoding is a 5×5 soft-argmax around the hard argmax, border-clamped and renormalised — in pipeline code, deliberately outside the exported graph.

Measured, and what it does NOT support

The A-CE ablation replaces the graded target with strict one-hot BCE. It did **not** fail — it **beat** the reference on aggregate (99.66% vs 98.96% m04 at matched steps), and the per-regime sweep shows the trade:

regime	reference (Gaussian)	A-CE (one-hot)
FAR ($s = 16$)	93.98	98.72 (+4.74)
VERY CLOSE ($s = 128$)	99.20	95.14 (-4.06)

The mechanism was measured directly (n=830 corners): at $\sigma = 2$, $\beta = 4$ the trained peaks have RMS radius **1.0028 px** with only **26.1%** of local probability mass on the peak pixel, against **0.2665 px** and **93.5%** for the one-hot arm — and the one-hot arm reaches that at 9k steps against a fully-trained 50k reference.

So the loss is asking for a blurry peak and getting one. The defensible claim is not “the Gaussian target is better”, but that σ and β are **task-appropriate hyperparameters that were inherited untuned** — $\beta = 4$ is CornerNet’s value for sparse object centres, not for a dense regular lattice. The `abl_sigma1` arm ($\sigma_{\text{hm}} = 1.0$, one key) tests whether tightening the target recovers the sharpness while keeping the graded target’s advantages.